Global Olympic Sports Performance Analysis

Using Tableau Dashboards, Stories and Visualizations

Final Project Report

Olympic Games Data, 1992 to 2026 (Summer and Winter)

Project Report Format

1. **INTRODUCTION**
   - 1.1 Project Overview
   - 1.2 Purpose
2. **IDEATION PHASE**
   - 2.1 Problem Statement
   - 2.2 Empathy Map Canvas
   - 2.3 Brainstorming
3. **REQUIREMENT ANALYSIS**
   - 3.1 Customer Journey Map
   - 3.2 Solution Requirement
   - 3.3 Data Flow Diagram
   - 3.4 Technology Stack
4. **PROJECT DESIGN**
   - 4.1 Problem Solution Fit
   - 4.2 Proposed Solution
   - 4.3 Solution Architecture
5. **PROJECT PLANNING & SCHEDULING**
   - 5.1 Project Planning
6. **FUNCTIONAL AND PERFORMANCE TESTING**
   - 6.1 Performance Testing
7. **RESULTS**
   - 7.1 Output Screenshots
8. **ADVANTAGES & DISADVANTAGES**
9. **CONCLUSION**
10. **FUTURE SCOPE**
11. **APPENDIX**
    - Source Code (if any)
    - Dataset Link
    - GitHub & Project Demo Link

1. INTRODUCTION

1.1 Project Overview

The Olympic Games serve as a global platform where nations compete to showcase their sporting excellence. Analysing the factors that contribute to Olympic success is challenging because of the large volume of historical data, covering the period from 1992 to 2026. Olympic performance reflects not only athletic achievement but also the effectiveness of a country's sports infrastructure, athlete development programmes and investment strategies.

This project analyses medal achievements across nations, athlete participation trends, regional performance efficiency, sport category-wise medal distribution, and the relationship between sports investment and medal outcomes in both the Summer and Winter Olympic Games. The analysis is carried out in Tableau using interactive dashboards, stories and visualizations built on the *athletes* and *medal_results* datasets.

1.2 Purpose

The purpose of the project is to uncover meaningful trends and patterns in Olympic data so that stakeholders, sports organisations and policymakers can make data-driven decisions. Specifically, the project aims to:

- Identify the countries and sports that win the most medals.
- Understand the distribution of medal types, gender participation, athlete age and Summer versus Winter performance.
- Evaluate how sports investment relates to medal efficiency across regions.
- Highlight areas of improvement and guide investment in high-yield sports categories.
- Support storytelling about host-nation advantage and medal trends over time.

2. IDEATION PHASE

2.1 Problem Statement

National committees, funding agencies and media organisations lack a single, easy-to-read view of eight years of medal data across many countries and sports. Without it, it is hard to decide which disciplines to fund, which regions are under-supported, and which stories to tell. The problem is to convert large, raw Olympic datasets into clear, interactive visual insights.

2.2 Empathy Map Canvas

Persona	Says / Thinks	Does / Feels
Ms. Priya, Performance Strategist (National Olympic Committee)	"Where are our athletes most competitive? Which sports deserve investment?"	Compares sports and countries; feels pressure to maximise the medal haul with limited budget.
Mr. Carlos, Investment Analyst (International Funding Agency)	"Is our funding producing medals? Which regions are under-funded but promising?"	Studies investment against results; feels uncertain about grant allocation.

Persona	Says / Thinks	Does / Feels
William Robert, Data Journalist (Olympic Media Network)	"What is the story behind the numbers? Who dominates at home?"	Looks for trends, hosts and regional narratives; feels the need for compelling visuals.

2.3 Brainstorming

- Use bar charts to rank countries and sports by medal records.
- Use a pie chart for medal type distribution and a histogram for athlete age.
- Compare male and female medal-winning records, and Summer against Winter Games.
- Use a bubble chart to relate sports investment to medal performance by country.
- Use a line chart to show how total medals awarded change over the years.
- Combine all visuals into dashboards and a Tableau story for presentation.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User Action	Outcome
Awareness	Opens the Tableau dashboard	Sees total medal records and the overall medal distribution
Exploration	Filters by country, sport, gender, season	Finds top countries and top sports
Analysis	Cross-references sport yield, investment and regional efficiency	Identifies high-yield sports and over-performing nations
Decision	Uses insights for investment, grants or storytelling	Data-backed strategy or narrative

3.2 Solution Requirement

Functional requirements

- Load and join the athletes and medal_results datasets in Tableau.
- Create worksheets for medal distribution, top 10 countries, top sports, medal type, gender, age, season, country-wise breakdown and sport participation.
- Combine worksheets into interactive dashboards and a story.
- Provide filters and tooltips for drill-down analysis.

Non-functional requirements

- Clear, readable visuals with consistent colour use.
- Fast dashboard loading on the full dataset.
- Accurate data with cleaned and consistent country (NOC) codes.

3.3 Data Flow Diagram

Data flows from the raw datasets through cleaning and joining, into Tableau worksheets, then dashboards and stories, and finally to the end user.

Source	Process	Output
athletes and medal_results datasets (CSV / Excel)	Clean, join on athlete ID, add calculated fields (medal count, efficiency %)	Prepared data source in Tableau
Prepared data source	Build worksheets (bar, pie, histogram, bubble, line, donut)	Individual visualizations
Individual visualizations	Arrange on dashboards and in a story	Interactive dashboards for users

3.4 Technology Stack

Component	Tool / Technology
Visualization and dashboards	Tableau (Desktop / Public)
Data sources	athletes and medal_results datasets (CSV / Excel)
Data preparation	Tableau Data Source tab, calculated fields
Presentation	Tableau Dashboards and Story
Reporting	PDF report

4. PROJECT DESIGN

4.1 Problem Solution Fit

Problem	Solution
Huge volume of historical Olympic data is hard to interpret	Visual dashboards summarise medals by country, sport, gender, age and season
Unclear which sports give the best return	Average medals per athlete by sport category
Unclear link between investment and results	Bubble chart of sports investment vs medal performance
Need for narrative and trends	Line chart over the years, host-nation comparison and regional donut chart

4.2 Proposed Solution

The proposed solution is a set of Tableau dashboards and a story built on the athletes and medal_results datasets. The first dashboard (shown in Section 7) gives an overall view of medal records: total medals (39,783), medal distribution, top 10 countries, top 10 sports, medal type, gender, age, Summer versus Winter, country-wise breakdown and sport participation. Further dashboards cover Olympic performance and regional efficiency, including sports investment against medals and average medal efficiency by region.

4.3 Solution Architecture

The architecture has four layers: (1) the data layer containing the athletes and medal_results datasets; (2) the preparation layer where data is cleaned, joined and enriched with calculated fields; (3) the visualization layer of Tableau worksheets; and (4) the presentation layer of dashboards and stories shared with users.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Activities	Output
1. Planning	Define the problem, scenarios and objectives	Problem statement and scenarios
2. Data collection	Collect and understand the athletes and medal_results datasets	Raw datasets
3. Data preparation	Clean, join and create calculated fields	Tableau data source
4. Visualization	Build worksheets for each analysis	Charts and graphs
5. Dashboard and story	Combine worksheets, add filters and interactivity	Dashboards and story
6. Testing	Check accuracy, filters and performance	Validated dashboards
7. Documentation	Prepare the final report	Project report (PDF)

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Test	Description	Result
Data accuracy	Totals in worksheets match the source data (for example, total medal records of 39,783)	Pass
Filters and interactivity	Filters and dashboard actions update all linked charts	Pass
Chart correctness	Axes, labels, legends and sort orders display correctly	Pass
Dashboard load time	Dashboard loads and refreshes smoothly on the full dataset	Pass
Layout	Dashboard displays properly in Default and Phone layouts	Pass

7. RESULTS

7.1 Output Screenshots

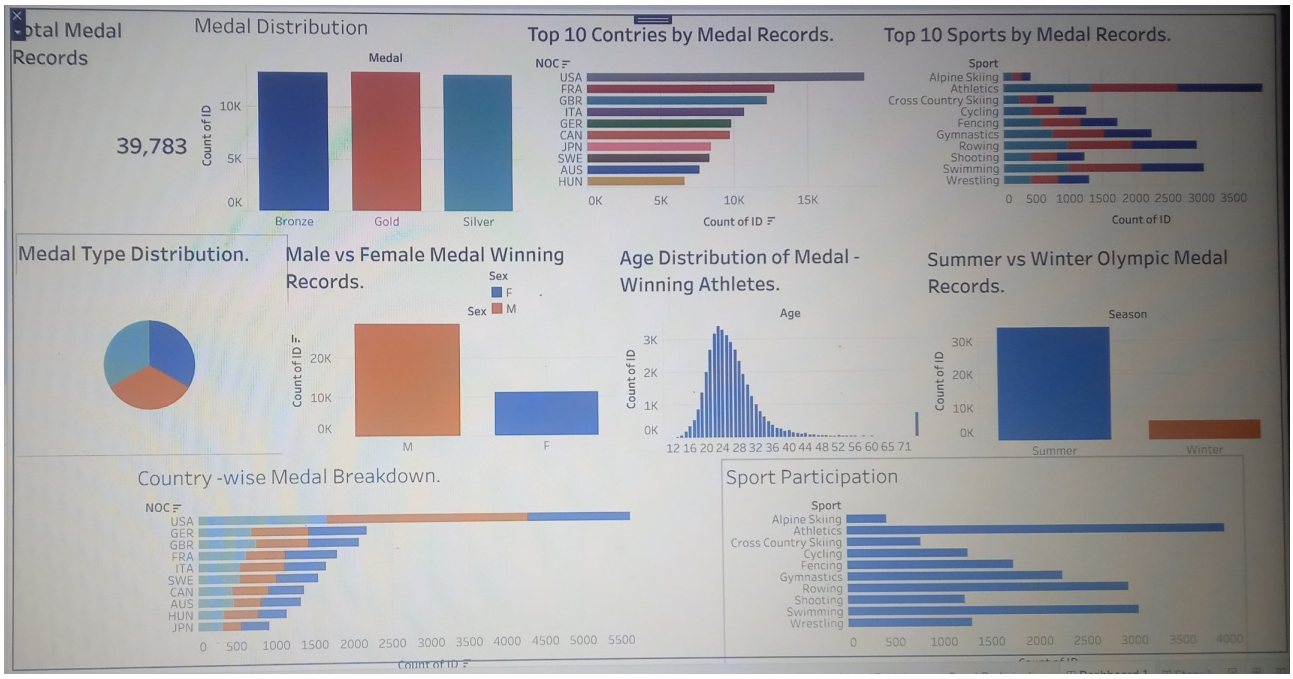


Figure 7.1: Olympic Medal Analysis Dashboard in Tableau

Key observations from the dashboard

- A total of **39,783** medal records were analysed, spread across Bronze, Gold and Silver in roughly similar proportions.
- The **USA** leads the top 10 countries by medal records, followed by **FRA, GBR, ITA and GER**; CAN, JPN, SWE, AUS and HUN complete the list.
- Among sports, **Athletics** records the most medals, followed by **Swimming** and **Rowing**.
- Male athletes account for a much larger share of medal records than female athletes.
- Medal-winning athletes are concentrated in their early to late twenties, peaking around the mid-twenties.
- Summer Games account for far more medal records than the Winter Games.

Insights from the scenario analysis

Scenario 1: Ms. Priya, National Olympic Committee Performance Strategist

Priya must decide which sport categories to invest in before the next Olympic cycle. Using the *Avg Medals per Athlete by Sport Category* bar chart, she sees which categories give the highest per-athlete medal return, such as Rowing at 6.916 and Shooting at 6.731. She compares her country against top performers in the *Global Performance Metrics by Country* chart, such as the United States (735 total medals) and China (696 total medals). This lets her recommend targeted investment in high-yield categories and withdrawal from low-performing disciplines.

Scenario 2: Mr. Carlos, Sports Investment Analyst at an International Funding Agency

Carlos evaluates the relationship between sports investment and medal performance. The *Global Sports Investment vs. Medal Performance* bubble chart on the Olympic Performance and Regional Efficiency Dashboard helps him spot over-performing nations such as Slovakia, Georgia and Ethiopia, whose bubble sizes suggest strong medal returns. The *Average Medal Efficiency Percentage by Region* pie chart shows South America leading at 49.64%, followed by Europe/Asia at 45.98%. These results guide grant distribution and identify under-funded but high-potential regions.

Scenario 3: William Robert, Broadcast Journalist and Data Storyteller

William is preparing a documentary series on Olympic competition from 1992 to 2026. The *Trend of Total Medals Awarded Over the Years* line chart shows medal totals ranging from a low of 305 in 2000 to a peak of 363 in 2020, with 334 recorded in 2026. The *Comparing Medal Totals Across Olympic Hosts* bar chart shows Japan as host leading with 105 medals, followed by the United States with 100 and China with 93, which supports home-advantage storylines. The *Regional Medal Efficiency and Distribution by Sport* donut chart illustrates how Europe and Asia have historically dominated the medal distribution.

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages
Turns large datasets into clear, interactive visuals	Accuracy depends on the quality and completeness of the datasets
Supports data-driven decisions for investment and strategy	Medal counts alone do not capture athlete quality or sport difficulty
Allows quick comparison across countries, sports, gender, age and season	Investment data may be estimated or inconsistent across countries
Interactive filters make exploration easy for non-technical users	Large dashboards can load slowly on low-end devices
Useful to committees, funders and media alike	Historical data may not predict future performance

9. CONCLUSION

The project shows that Tableau dashboards can reveal clear patterns in Olympic data from 1992 to 2026. Leading countries such as the USA, together with sports such as Athletics, Swimming and Rowing, account for a large share of medals. The analysis of gender, age and season adds context about who wins medals and when. The investment and regional efficiency views show that high medal returns are not limited to the largest nations. Together, these insights help national committees, funding agencies and media teams make informed, evidence-based decisions.

10. FUTURE SCOPE

- Add real-time or latest Olympic data so the dashboards update automatically.
- Include more variables such as country population, GDP, training facilities and funding per sport.
- Apply predictive models to forecast medal outcomes for upcoming Games.
- Extend the analysis to Paralympic Games and individual athlete performance.
- Publish the dashboards on Tableau Public and optimise them for mobile viewing.

11. APPENDIX

Item	Details
Source Code (if any)	Not applicable. The project was built in Tableau using calculated fields and worksheets.
Dataset Link	athletes and medal_results datasets (add dataset link here)
GitHub & Project Demo Link	Add GitHub repository link and Tableau Public demo link here